

Reachability-Certified Autonomy for On-Orbit Docking and Servicing

1st Santiago Thorup

Aeronautics & Astronautics - Stanford University

AA 276 - Safety Critical Autonomy

Stanford, United States of America

sthorup4@stanford.edu

Abstract—This work presents a Hamilton–Jacobi (HJ) reachability-based framework for provably safe autonomous on-orbit docking under worst-case model uncertainty. We formulate planar docking as a six-dimensional reach-avoid differential game in the target-centered Clohessy–Wiltshire frame, whose solution yields a Backward Reachable Tube (BRT) certifying all initial states from which the chaser can reach the docking port while avoiding the target’s body. To overcome the curse of dimensionality, I decompose the 6-D problem into a 4-D translational BRT (position/velocity) and a 2-D attitude BRT (orientation/rate), each solved efficiently on a JAX-based ENO grid. From the resulting value functions we extract a disturbance-robust feedback policy and compare it, via closed-loop Monte Carlo simulations, against three fuel-aware safety filters: a simple nominal LQR, a least-restrictive safety filter, and a QP based smooth-blending filter. Results over randomized initial conditions confirm that pure robust control guarantees 100% safety at high propellant cost, LQR alone minimizes fuel but fails to avoid collisions, and the proposed switching and blending filters achieve formal safety with reduced fuel consumption. This study demonstrates the viability of HJ reachability for complex, safety-critical space maneuvers and lays the groundwork for extension to full 3-D dynamics via MPC-assisted neural solvers.

I. INTRODUCTION AND MOTIVATION

On-orbit servicing, refueling, and debris-removal are rapidly transitioning from fiction to reality, opening a wide array of commercial and governmental applications. Many forthcoming large-scale spacecrafts would be infeasible—or prohibitively expensive—without reliable in-situ servicing. NASA’s proposed Habitable Worlds Observatory (HWO) exemplifies this trend: its sheer size requires autonomous on-orbit assembly and periodic maintenance, and its estimated \$11 billion life-cycle cost would make the platform an irreplaceable national asset [1]. As a result, **guaranteed safety during fully autonomous rendezvous and docking** has become a hard requirement rather than a desirable bonus. Yet existing rendezvous algorithms like LQR managers, pulse-based MPC, and vision-guided GN&C pipelines deliver high performance but no formal safety guarantees.

To close this gap the present work casts planar docking as a two-player Hamilton–Jacobi (HJ) reach-avoid differential game. Solving the resulting partial differential equation (PDE) yields a certified Backward Reachable Tube (BRT) whose zero-level set encloses every initial condition from which the chaser can reach the docking manifold while avoiding the target body under worst-case model uncertainty. The

obtained value function allows for the synthesis of a real-time feedback policy that keeps the vehicle inside this safe corridor. By delivering end-to-end mathematically grounded safety assurances, the approach lays essential groundwork for trustworthy autonomous servicing, life-extension, and debris-mitigation missions.

II. RELATED WORK

This project is not based off any previous work and is informed by current research initiatives at NASA’s Jet Propulsion Laboratory (JPL). However, Hamilton–Jacobi (HJ) reachability has a small but fast-growing body of literature in spacecraft proximity operations as outlined below.

- *Applied Reachability Analysis for Spacecraft Rendezvous and Docking with a Tumbling Object* [2].
 - Extends HJ reachability to docking with a rotating target.
- *Computing reach-avoid sets for space vehicle docking under continuous thrust* [3].
 - Solves 4D - CW dynamics with continuous-thrust inputs, producing BRT value function with minimum-fuel/minimum time policies.
- *Under-Approximating Reach-Avoid Sets for Space Vehicle Maneuvering in the Presence of Debris* [4].
 - Solves 6D - relative orbit model with multiple keep-out (debris) zones during a docking scenario.

For my formulation I will utilize two open source software packages to help solve my PDE which have been outlined below.

- **Grid-based solvers.** Stanford ASL Labs open-source `hj_reachability` toolbox implements an ENO scheme suitable for 6-D problems [5].
- **Neural approximations.** DeepReach make HJ analysis in higher dimensions tractable and will be explored for a possible 3-D extension [6].

For boundary conditions and docking tolerances we adopt the NASA International Docking System Standard Interface Definition Document [7], and for planar orbital dynamics we rely on the canonical Clohessy–Wiltshire equations [8]. Collectively, these references provide the theoretical, numerical, and empirical groundwork upon which the present 6D reach-avoid formulation is built.

III. PROBLEM FORMULATION

A. State Definition

Let the relative state between the chaser spacecraft $(\cdot)_c$ and target spacecraft $(\cdot)_t$ be

$$x = [p_x \ p_y \ v_x \ v_y \ \theta \ \omega]^\top \in \mathbb{R}^6,$$

The translational components $p_{x,y} = p_{x,y,c} - p_{x,y,t}$ and $v_{x,y} = v_{x,y,c} - v_{x,y,t}$ are defined in the *target-centred Hill frame*. This frame is a Cartesian triad that *rotates* with the target's circular orbit: the $+x$ axis points radially outward from Earth, the $+y$ axis is aligned with the target's velocity vector, and the $+z$ axis completes the right-hand system (neglected for this planar implementation). As the frame translates and rotates with the target's center of mass, the target itself remains stationary with zero velocity in its coordinate frame. Therefore, the relative state $x = [p_x, p_y, v_x, v_y, \theta, \omega]^\top$ captures the chaser's motion *w.r.t the target*. The attitude variables $\theta = \theta_c - \theta_t$ and $\omega = \dot{\theta}_c - \dot{\theta}_t$ describe the angular offset between the target and chaser spacecraft and its rate. Working in this target-centred, rotating frame yields two benefits:

- The goal set becomes *static* in state space.
- For small separations the translational dynamics linearize to the time-invariant Clohessy–Wiltshire equations.

Control and disturbance: Controls $u := [u_x, u_y, u_\theta]^\top \in \mathcal{U}$ are bounded by $|u_x|, |u_y| \leq u_{t(max)}$ and $|u_\theta| \leq u_{\theta max}$. Model-uncertainty $d := [d_x, d_y, d_\theta]^\top \in \mathcal{D}$ bounded by $|d_x|, |d_y| \leq d_{max}$ and $|d_\theta| \leq d_{\theta max}$.

B. Chaser/Target Dynamics

We are defining our system dynamics in a target-centered frame as described in the state definition section above III-A, with $+\hat{x}$ radial and $+\hat{y}$ along-track. Therefore, linearizing the two-body equations about the target (assuming a near-circular orbit with a mean motion n) yields the planar Clohessy–Wiltshire (CW) translational dynamics $\ddot{p}_x = 3n^2 p_x + 2n \dot{p}_y + a_x$ and $\ddot{p}_y = -2n \dot{p}_x + a_y$ [8]. The additional terms (a_x, a_y) relate to the chaser's net accelerations and can be split into *control* (u) and *model-uncertainty* (d). The attitude of the system can be appended onto the dynamics using the following principal axis planar relations: $\dot{\theta} = \omega - d_\theta$, $\dot{\omega} = u_\theta/J_c$ [9]. Therefore, given the state $x = [p_x, p_y, v_x, v_y, \theta, \omega]^\top$, control $u = [u_x, u_y, u_\theta]^\top$, and disturbance $d = [d_x, d_y, d_\theta]^\top$, the resulting six-state, control-affine system can be defined as

$$\dot{x} = Ax + Bu + Ed, \quad (1)$$

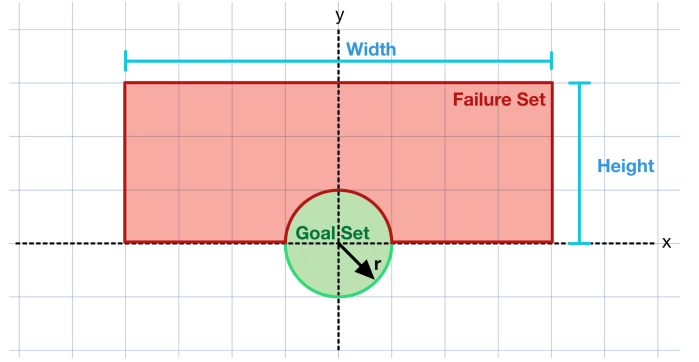


Fig. 1. Goal and Failure Set

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 3n^2 & 0 & 0 & 2n & 0 & 0 \\ 0 & 0 & -2n & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (2)$$

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{m_c} & 0 & 0 \\ 0 & \frac{1}{m_c} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{J_c} \end{bmatrix}, \quad E = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (3)$$

Where, m_c and J_c are the mass and planar moment of inertia of the chaser respectively. Matrix A embeds the CW coupling $\begin{bmatrix} 3n^2 & 2n \\ -2n & 0 \end{bmatrix}$; B scales thrust by m_c and torque by J_c ; E includes bounded uncertainties, resulting in a formulation compatible with Hamilton–Jacobi reachability analysis.

C. Goal and Failure Sets

Due to the relative dynamics of the system the goal set is stationary and centered at the origin as described in state definition section above III-A. Therefore, with given position, velocity, and attitude tolerances $\varepsilon_p, \varepsilon_v, \varepsilon_\theta, \varepsilon_\omega > 0$ the goal set can be defined as follows.

Goal set.

$$\mathcal{G} = \left\{ x \mid |p_{x,y}| \leq \varepsilon_p, |v_{x,y}| \leq \varepsilon_v, |\theta - \pi/2| \leq \varepsilon_\theta, |\omega| \leq \varepsilon_\omega \right\}.$$

Failure set geometry. We define the failure set as body of the target spacecraft. Let w (width), h (height), and r (docking indent radius) define the the failure set as shown in figure 1.

Define two signed-distance primitives:

$$S_{\text{rect}}(p_x, p_y) = \max\left\{ |p_x| - \frac{w}{2}, -p_y, p_y - h \right\}, \quad (4)$$

$$S_{\text{indent}}(p_x, p_y) = \max\left\{ p_y, \sqrt{p_x^2 + p_y^2} - r \right\}. \quad (5)$$

$S_{\text{rect}} \leq 0$ iff the point is inside the spacecraft (rectangle), $S_{\text{indent}} \leq 0$ iff the point is inside the downward-facing semicircle. The target-body failure set is therefore

$$g(p_x, p_y) = \max\{S_{\text{rect}}(p_x, p_y), -S_{\text{indent}}(p_x, p_y)\}, \quad (6)$$

$$\mathcal{F} = \{x \in \mathbb{R}^6 \mid g(p_x, p_y) \leq 0\}. \quad (7)$$

Figure 1 depicts both $\mathcal{G}$ (green) and $\mathcal{F}$ (red).

D. Reach-Avoid Formulation

Define signed-distance functions ℓ and g such that $\ell(x) \leq 0 \Leftrightarrow x \in \mathcal{G}$ (goal) and $g(x) \leq 0 \Leftrightarrow x \in \mathcal{F}$ (failure). For a finite horizon T the set of initial states that can reach $\mathcal{G}$ while avoiding $\mathcal{F}$ under worst case disturbances is defined as the *Backward-reachable tube (BRT)*.

$$\begin{aligned} \text{BRT}(t) = \left\{ x \mid \forall d(\cdot) \in \mathcal{D}^{[t,T]}, \exists u(\cdot) \in \mathcal{U}^{[t,T]}, \right. \\ \left. \xi_{x,t}^{u,d}(\tau_1) \in \mathcal{G} \text{ for some } \tau_1 \in [t,T] \right. \\ \left. \cap \xi_{x,t}^{u,d}(\tau_2) \notin \mathcal{F} \forall \tau_2 \in [t,T] \right\}. \end{aligned} \quad (8)$$

Given this formulation, you can define the cost function 9 as follows: ensuring you enter $\mathcal{G}$ while avoiding $\mathcal{F}$.

Cost Function.

$$J(x, u(\cdot), d(\cdot), t) = \max \left\{ \min_{\tau_1 \in [t,T]} \ell(x^{u,d}(\tau_1)), \right. \\ \left. \max_{\tau_2 \in [t,T]} -g(x^{u,d}(\tau_2)) \right\}. \quad (9)$$

From 9 you define the value function 10 which yields the PDE 11.

Value function and HJ-Isaacs PDE.

$$V(x, t) = \min_{u(\cdot)} \max_{d(\cdot)} J(x, u(\cdot), d(\cdot), t), \quad (10)$$

$$\begin{aligned} \max \left\{ \min \left\{ \frac{\partial V}{\partial t} + \min_{u \in \mathcal{U}} \max_{d \in \mathcal{D}} \frac{\partial V}{\partial x} f(x, u, d), \ell(x) - V(x, t) \right\}, \right. \\ \left. -g(x) - V(x, t) \right\} = 0, \quad t \in [0, T], \end{aligned} \quad (11)$$

with terminal condition $V(x, T) = \max\{\ell(x), -g(x)\}$.

Solving the PDE 11 using the proposed boundary condition yields the desired value function which is used in controller synthesis.

Robust feedback controller.

$$u^*(x, t) = \arg \min_{u \in \mathcal{U}} \max_{d \in \mathcal{D}} \frac{\partial V}{\partial x}(x, t) \cdot f(x, u, d) \quad (12)$$

Therefore, any trajectory starting in $\{x \mid V(x, 0) \leq 0\}$ and controlled by (12) is guaranteed to reach $\mathcal{G}$ without entering $\mathcal{F}$ within horizon T .

E. Safety Filtering

In addition to guaranteeing collision avoidance, we must also account for fuel use. A purely safety-optimal feedback law (solving the HJI minimax at every state, see equation 12) can be overly conservative and exhaust thruster resources. To strike a balance, we introduce a *safety filter* that combines a fuel-efficient “nominal” controller with the HJ-derived robust controller.

Nominal LQR Control: In order to define a fuel-efficient nominal controller we define a continuous-time LQR law on the full six-state linearized CW dynamics III-B (neglecting the failure set and disturbance). Let

$$x = [p_x \ p_y \ v_x \ v_y \ \theta \ \omega]^\top, \quad u_{\text{nom}}(x) = \begin{bmatrix} u_x \\ u_y \\ u_\theta \end{bmatrix}.$$

Given these dynamics we must define two symmetric, positive-definite weight matrices

$$Q = \text{diag}(q_1, q_2, q_3, q_4, q_5, q_6), \quad R = \text{diag}(r_1, r_2, r_3).$$

Where Q is the state cost matrix and R is the control cost matrix. Given these two matrices we define a cost function which is minimized over the trajectory.

$$J(x) = (x^\top Q x + u^\top R u) d\tau$$

Then the continuous-algebraic Riccati equation (CARE)

$$A^\top P + P A - P B R^{-1} B^\top P + Q = 0 \quad (13)$$

is solved for $P = P^\top \succ 0$. The resulting constant gain matrix

$$K = R^{-1} B^\top P$$

which yields the nominal state-feedback law

$$u_{\text{nom}}(x) = -K(x - x_d), \quad (14)$$

where $x_d = [0, 0, 0, 0, \theta_{\text{target}}, 0]^\top$ is the docking-port target state (for planar docking, $\theta_{\text{target}} = \frac{\pi}{2}$). In practice we clip each component of u_{nom} to its respective actuator limit:

$$u_{i,\min} \leq [u_{\text{nom}}]_i \leq u_{i,\max}, \quad i = 1, 2, 3,$$

so that $\|u_{\text{nom}}\|$ never exceeds the maximum thrust $\bar{u}_t$ or torque $\bar{u}_\theta$.

The choice of Q and R (choose $R \gg Q$) drive the chaser spacecraft toward the docking target while minimizing fuel use. However, it is important to note that u_{nom} has no knowledge of the failure set $\mathcal{F}$ and therefore may steer the spacecraft directly through the target body if unconstrained. This motivates the use of a safety-filter overlay.

Least-Restrictive (Switching) Filter: Let $V(x, t)$ denote the reach-avoid value function from Section III-D, whose zero sub-level set $V(x, t) \leq 0$ is the certified safe set at time t from which there exists a control input that can drive the system to the target set within the time horizon T . Given this definition we can define the *least-restrictive* filter as a simple switch:

$$u(x, t) = \begin{cases} u_{\text{nom}}(x), & \text{if } V(x, t) < 0, \\ u^*(x, t), & \text{if } V(x, t) \geq 0, \end{cases} \quad (15)$$

where $u^*(x, t)$ is the safety-optimal feedback policy from (12). In other words, whenever the state is strictly inside the BRT ($V < 0$), we apply the fuel-efficient LQR command; once the trajectory reaches the zero-level boundary ($V = 0$), we switch to the robust u^* to ensure the state remains within the BRT and therefore safe.

Smooth Blending Filter: A hard switch can introduce discontinuities in the commanded thrust/torque and may cause chattering near the BRT boundary ($V = 0$). This control law may also favor the robust safe control $u^*(x, t)$ if the systems initial state is near the BRT boundary leading to sub-optimal fuel usage. To combat these issues we can implement a *QP-based smooth blending filter* that attempts to find a control input u that satisfies the safety condition derived from the HJ PDE while minimizing its $L2_{norm}$ distance from the fuel efficient control u_{nom} .

Let $\nabla_x V(x, t)$ be the spatial gradient of the value function at (x, t) , and denote the system dynamics by $\dot{x} = f(x, u, d)$. We require that the worst-case Lie derivative of V along f is nonpositive near the boundary. Specifically, for a small constant $\gamma > 0$, we enforce

$$\min_{d \in \mathcal{D}} [\nabla_x V(x, t) \cdot f(x, u, d)] \geq -\gamma V(x, t). \quad (16)$$

Because $V(x, t) \leq 0$ inside the BRT and $-\gamma V(x, t) \geq 0$ when $V \geq 0$, inequality (16) guarantees that V cannot cross from negative to positive (the state cannot exit the BRT) under the chosen u .

Thus, the *smooth blending filter* at state-time (x, t) is given by the quadratic program

$$\begin{aligned} u^*(x, t) &= \arg \min_{u \in \mathcal{U}} \|u - u_{nom}(x)\|_2^2 \\ \text{s.t. } \min_{d \in \mathcal{D}} \{ \nabla_x V(x, t) \cdot f(x, u, d) \} &\geq -\gamma V(x, t). \end{aligned} \quad (17)$$

However, given the importance of fuel consumption for space based missions, the optimal feedback control law 12 will be

IV. APPROACH

A. Reach-Avoid Formulation

Traditional safety oriented spacecraft GN&C pipelines split “guidance” (a nominal trajectory) from a downstream “safety filter” (e.g. CBF or BRT) that intervenes whenever the system is in danger. Instead posing docking as a single *reach-avoid differential game*, safety and goal-achievement are combined in the same Hamilton–Jacobi (HJ) value function (§III-D), eliminating the need for an nominal controller while preserving formal safety guarantees under worst case disturbances. The fact that additional mathematical complexity is minimal as the obstacle terms ℓ and g enter the HJI PDE as simple “max/min” operators make this formulation ideal.

However, we must also manage fuel consumption. A purely safety-optimal feedback law $u^*(x, t)$ (Eq. (12)) can be overly conservative and quickly exhaust thruster resources. To address this, we layer a *safety filter* (see §III-E) on top of a fuel-efficient nominal controller, blending the two as needed so that we maintain safety while minimizing integrated thrust.

B. Grid-Based Solver

A certifiable baseline can be found using the HJI PDE grid based solver `hj_reachability`. This open source

tool box utilizes a grid to produce a monotone approximation with provable error bounds, providing the “ground-truth” value function $V_{grid}(x, t)$. Due to the curse of dimensionality, the upper limit for grid based solvers is 6D justifying the (planar translation and single axis rotation) model defined in the dynamics section above III-B. However, even at 6 dimensions, the vanilla grid solver can become computationally prohibitive. In preliminary tests, solving a $41 \times 41 \times 31 \times 31 \times 31 \times 31$ grid took multiple hours of GPU time and near-capacity VRAM. To mitigate this, I split the problem into two smaller reach-avoid subproblems as the CW translational and rotational dynamics are not coupled:

- 1) A *4-D translational BRT* on (p_x, p_y, v_x, v_y) with the geometry-only failure SDF $g_{tr}(p_x, p_y)$ from §III-C, and the translational goal SDF $\ell_{tr}(p_x, p_y, v_x, v_y)$. The corresponding dynamics are the CW equations with control $u = [u_x, u_y]$ and disturbance $[d_x, d_y]$.
- 2) A *2-D attitude BRT* on (θ, ω) with goal SDF $\ell_{att}(\theta, \omega)$ and (optionally) a trivial or empty failure set. The dynamics are $\dot{\theta} = \omega - d_\theta$, $\dot{\omega} = \frac{1}{J_c} u_\theta$.

Each subproblem is solved independently using `hj_reachability.solve` on a dedicated grid (e.g. $51 \times 51 \times 31 \times 31$ for translation and 361×141 for attitude). The two value functions $V_{tr}(p_x, p_y, v_x, v_y, t)$ and $V_{att}(\theta, \omega, t)$ are then used independently to derive the optimal safe control policy at every state and time.

$$\mathcal{R}(0) = \{x \mid V_{tr}(p_x, p_y, v_x, v_y, 0) \leq 0 \wedge V_{att}(\theta, \omega, 0) \leq 0\}.$$

This splitting reduces memory usage by an order of magnitude, cuts solve time from hours to minutes, as allows for increased grid resolution at the cost of mild conservatism.

C. Controller Synthesis and Verification

Once the grid-based value functions V_{tr} and V_{att} are available, we extract disturbance-robust feedback laws by evaluating

$$\begin{aligned} u_{tr}^*(x_{tr}, t) &= \arg \min_{u_{x,y}} \max_{d_{x,y}} \nabla_{x_{tr}} V_{tr} \cdot f_{tr}(x_{tr}, u_{x,y}, d_{x,y}), \\ u_{att}^*(x_{att}, t) &= \arg \min_{u_\theta} \max_{d_\theta} \nabla_{x_{att}} V_{att} \cdot f_{att}(x_{att}, u_\theta, d_\theta). \end{aligned}$$

In closed-loop Monte-Carlo simulations we compare several safety-filtering strategies:

- **Baseline (No Filter):** Apply the nominal LQR control $u_{nom}(x)$ for all t .
- **Switching Filter (Eq. (15)):** Use u_{nom} when $V(x, t) < 0$, switch to $u^*(x, t)$ when $V(x, t) \geq 0$.
- **QP Smooth Blending (Eq. (17)):** Solve the QP at each time step to find the closest u to u_{nom} that satisfies the safety-constraint $\min_d \{ \nabla_x V \cdot f \} \geq -\gamma V(x, t)$.
- **Pure Robust:** Always apply $u^*(x, t)$ (no nominal controller).

We evaluate each strategy according to:

- **Safety Rate:** Fraction of trials that avoid $\mathcal{F}$ for all $t \in [0, T]$.

- **Docking Success:** Fraction of trials that reach $\mathcal{G}$ within the horizon T .
- **Fuel Usage:** Approximated by

$$\int_{t_0}^{t_{\text{dock}}} \|u(t)\|_1 dt,$$

where we assume a fixed specific impulse $I_{sp} = 300$ s and a 1-Hz sampling of the commanded control. The thrust levels translate to propellant mass via

$$\Delta m = \frac{|T|}{I_{sp} g_0} \Delta t,$$

summing over each 1-Hz step until docking.

V. RESULTS AND ANALYSIS

A. 2D Attitude BRT

We first compute the 2D reach-avoid tube (BRT) in the (θ, ω) subspace, where θ is the chaser's relative angular displacement with respect to the target docking port and ω is the relative angular rate. The goal set for this BRT is

$$\{|\theta - \pi/2| \leq \varepsilon_\theta, |\omega| \leq \varepsilon_\omega\}.$$

i.e. aligned with the docking interface and zero angular rate. Since orientations outside this “capture angle” do not constitute a collision in the planar model, we do not include any failure set for the 2D problem.

Figure 2 shows a slice of $V_{\text{att}}(\theta, \omega, t)$ at $t = 25$ s. All (θ, ω) pairs lying in the shaded region with $V_{\text{att}} \leq 0$ can reach the target set within the 25s timeframe despite worst-case disturbance d_θ . Notice that:

- States with large $|\theta - \pi/2|$ (far from alignment) must also have very small $|\omega|$ to be inside the BRT—otherwise the chaser cannot brake its tumble in time.
- Conversely, states near $\theta = \pi/2$ can tolerate a wider range of ω (up to approximately ± 0.6 rad/s) because only a small attitude correction is required.
- The “slanted” shape of the zero-level contour reflects the coupling between attitude error and rotation rate: a larger angular error can still be corrected if the rate is pointing toward the goal (negative ω if $\theta > \pi/2$, positive ω if $\theta < \pi/2$).

B. 4D Translational BRT

Next, we solve the 4D reach-avoid BRT in the translation subspace (p_x, p_y, v_x, v_y) . The target (goal) set is

$$\mathcal{G} = \{x \mid |p_{x,y}| \leq \varepsilon_p, |v_{x,y}| \leq \varepsilon_v, \}$$

i.e. zero relative position and zero relative velocity at the docking port. The failure set $\mathcal{F}$ is defined by the rectangular body of the tumbling target (with downward-facing semicircular indent) as show in section III-C. All states inside that geometry are automatically unsafe because they correspond to collision with the target's body.

Figure 3 displays a 2D slice of the 4D value function $V_{\text{tr}}(p_x, p_y, v_x = 0, v_y = 0, t)$ at $t = -25$ s. That is, we fix

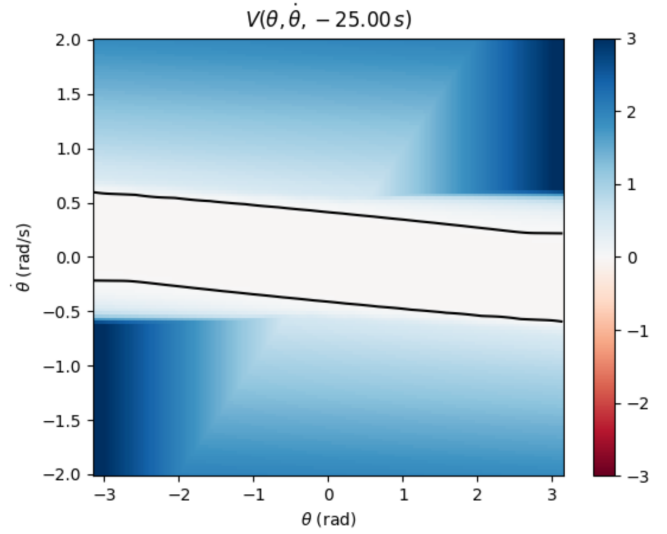


Fig. 2. 2D Rotational BRT at $t = -25$ s. The shaded region ($V_{\text{att}} \leq 0$) shows all (θ, ω) from which alignment $\theta = \pi/2$, $\omega = 0$ is guaranteed within 25s under worst-case d_θ .

$(v_x, v_y) = (0, 0)$ and visualize how the BRT (zero sublevel) lies in the (p_x, p_y) plane.

Key observations:

- The central “U-shaped” **red** region encloses all positions (with zero initial velocity) from which there exists a control that can drive the chaser into the docking indent and brake to zero velocity within 30s while avoiding the rectangular body.
- Immediately surrounding the indent, there is a thin white band (near the zero-level contour) that represents states requiring nearly maximal thrust to succeed. Any slight disturbance or delay for those points would push them out of the red region and into the blue, making docking impossible within the time horizon.
- Farther out—beyond roughly $|p_x| \approx 10$ m and/or $|p_y| \approx 7$ m—the plot turns **blue**, indicating that a spacecraft starting at rest too far from the dock cannot both skirt around the targets body and arrive at the docking port with zero velocity in 30s. Those blue states are *unreachable* under worst-case (d_x, d_y) .

C. Trajectories Under Different Control Laws

Finally, we simulate the full 6D closed-loop dynamics under four different control strategies, starting from the same initial condition x_0 within the BRT for both the 2D and 4D system. Figure 4 outlines 4 trajectories under these different control laws showing:

- 1) Three snapshots of the time-varying 4D value-function contour $V(x, t)$ over the (p_x, p_y) plane at $t = -30$ s, -15 s, 0 s. The chaser's position is depicted by the blue square, and the target body and dock indent are outlined in black.
- 2) The corresponding control-input time histories $\{u_x(t), u_y(t), u_\theta(t)\}$ over the 30-s simulation.

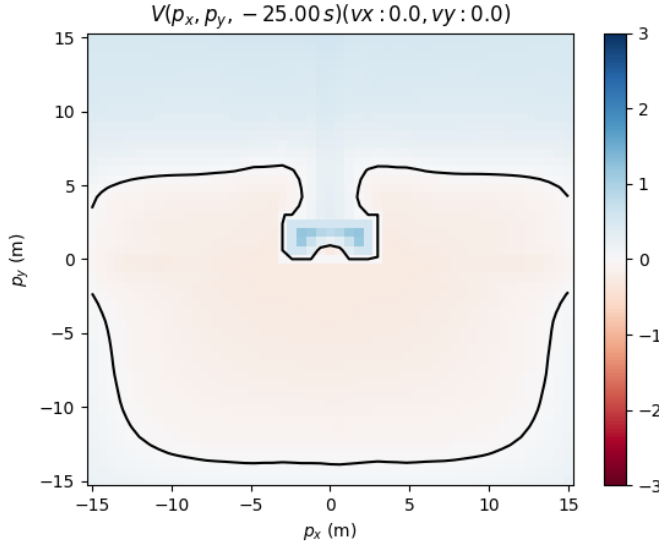


Fig. 3. 4D Translational BRT slice at $v_x = 0$, $v_y = 0$, $t = -25$ s. The zero-level contour (black) demarcates the boundary of states from which docking with zero velocity is guaranteed within 25s under worst-case (d_x, d_y) . Inside (red) is safe; outside (blue) is not.

Baseline (No Filter): Applying purely the LQR nominal control u_{nom} drives the spacecraft directly toward the docking port. However, even though the initial position is nominally within the 6D BRT slice, LQR does not avoid the rectangular body. In the middle snapshot ($t = -15$ s), the trajectory has intersected the keep-out zone which would lead to a collision with the target. The control plot shows a smooth ramp in u_x, u_y that simply minimizes fuel, but it never “knows” to avoid the failure set. At $t \approx 10$ s, the trajectory crosses into the failure set, leading to docking failure.

Least-Restrictive Switching Filter (Eq. (15)): The second row shows the same initial condition but with the switching safety filter:

While $V < 0$, use LQR.
Once $V \geq 0$, switch to u^* .

Initially, $V(x_0, -30) < 0$ but close enough to the boundary so the spacecraft uses the the robust feedback controller u^* . The value-function contours update as t advances, and near $t \approx 11$ s, the state becomes safe enough for the least restrictive filter to allow for fuel efficient LQR nominal control. Near $t \approx 14$ s the state approaches the BRT boundary again ($V \approx 0$) and the controller switches to u^* , guiding the chaser around the rectangular body. The control plot shows a large number of discrete thrust commands (red and green spikes) after the switch, reflecting the robust safety policy countering worst-case disturbance. Ultimately, the chaser arrives at $\mathcal{G}$ around $t \approx 27$ s with improved fuel usage compared to purely robust control. Safety and docking success are both achieved.

Pure Robust Controller u^* (Eq. (12)): The third row applies u^* at all times. From $t = 0$, the feedback attempts to keep $\argmin_u (\nabla V \cdot f(x, u, d)) < 0$, which necessarily means

applying near-maximal thrust and torque. In the snapshots, the chaser takes a wide path around the obstacle. The control plot is densely filled with high-magnitude spikes, and cumulative fuel usage is highest of all strategies. Although docking and collision avoidance is guaranteed, the fuel penalty is substantial.

Smooth Blending Filter (Eq. (17)): The fourth row uses the QP-based filter. This filter attempts to minimize the $L_{2\text{norm}}$ distance between a safe control input and the fuel efficient u_{nom} . This can be clearly seen when $t \approx 22$ s in the control plot as u_x, u_y control inputs almost drop to zero. However, since this smooth blending filter relaxes full robustness, the chaser’s final approach in the third snapshot deviates by a few centimeters from the exact dock position. This leads to guaranteed collision avoidance but unreliable docking success within 30s. However, fuel usage is reduced compared to both pure robust control and the least restrictive safety filter.

D. Monte Carlo Simulation Results

We performed 5 randomized trials for each of the four control strategies, sampling initial states uniformly in the same region and recording whether the trajectory entered the failure set $\mathcal{F}$, reached the docking goal $\mathcal{G}$, and docked safely (i.e. reached $\mathcal{G}$ without ever entering $\mathcal{F}$), as well as the propellant consumed. Table I summarizes these outcomes.

TABLE I
MONTE CARLO RESULTS OVER 5 TRIALS FOR EACH CONTROLLER

Controller	# Failure	# Reached	# Safe Dock	Avg. Fuel (kg)
Robust safety only	0	5	5	0.3046
Least-restrictive switch	0	5	5	0.2988
Nominal (no filter)	3	1	0	0.0763
Smooth blending (QP)	0	2	2	0.1811

Key observations:

- **Robust safety only** achieves 100% safety and 100% docking success, but at the highest average fuel (0.3046kg).
- **Least-restrictive switching** also maintains perfect safety and success, while reducing fuel slightly (0.2988kg) by using the nominal controller whenever safely possible.
- **Nominal LQR alone** uses the least propellant (0.0763kg) but fails in 60% of trials, colliding with the target body or never reaching the dock.
- **Smooth blending** trades off safety and efficiency: it avoids collisions in all trials (0 failures), but only succeeds in 40% of cases, with moderate fuel use (0.1811kg).

These results quantitatively confirm the trade-offs: pure robustness guarantees safety at high fuel cost; pure nominal control minimizes fuel but is unsafe; switching and smooth-blending filters lie in between, offering formal safety guarantees with substantially reduced propellant consumption.

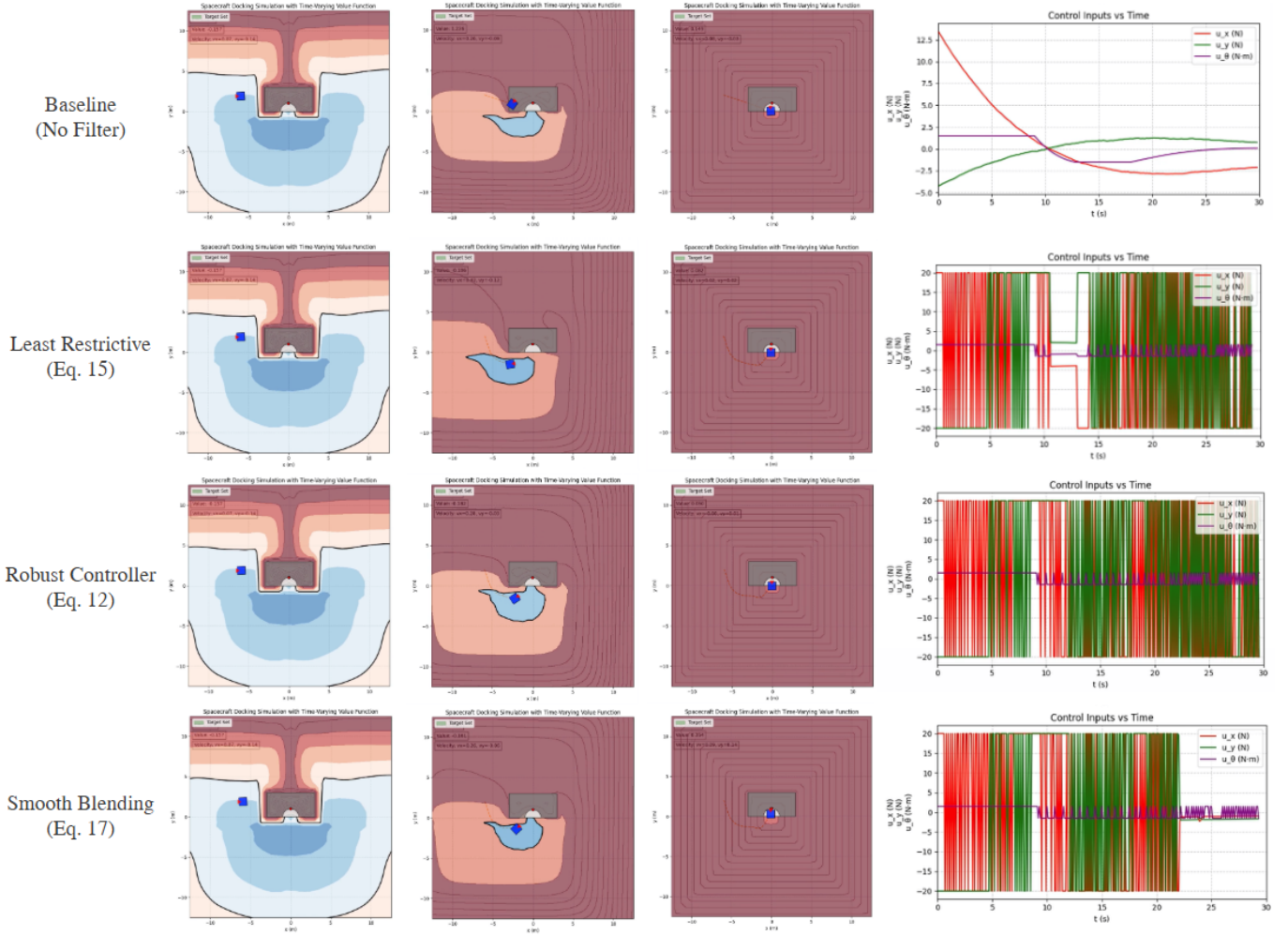


Fig. 4. Spacecraft docking simulations under four control laws. Each row shows three sequential frames of the translational value function $V_{tr}(p_x, p_y, t)$ (at $t = -30s, -15s$ and $0s$), overlaid with the chaser's position trajectory (blue squares) and the target-body geometry (black outline). On the right, the corresponding control time histories of u_x , u_y , and u_θ over the full 30s are depicted.

VI. FUTURE WORK

This work demonstrates that Hamilton–Jacobi reachability can be applied to complex, safety-critical tasks such as autonomous in-orbit docking. Moving forward, we plan to:

- **Extend to full 3-D dynamics.** Recast the reach–avoid formulation using realistic 6-DOF orbital and attitude dynamics, including gravitational perturbations and higher-fidelity thruster models.
- **Leverage MPC-assisted DeepReach.** Integrate a model-predictive control (MPC) guidance layer with the DeepReach physics-informed neural PDE solver to tractably compute reach–avoid tubes in the higher-dimensional setting.

Other possible areas of work include:

- Integrating a secondary fuel-optimal objective directly into the HJ formulation (multi-objective reachability).
- Modeling plume impingement zones as additional, state-dependent failure sets.

- Introducing dynamic debris fields (moving obstacles) and time-varying keep-out zones.
- Accommodating more complex coupled target-spacecraft disturbances (e.g. tumbling with uncertain inertia, mis-fired thrusters).
- Hardware-in-the-loop validation on a 6-D planar air-bearing testbed to assess real-time performance and robustness.

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